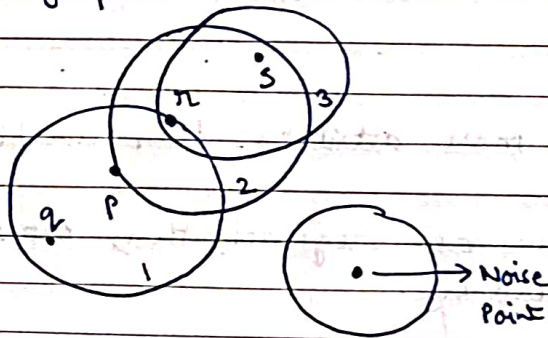


DBScan Cluster (Density Band Separate Cluster Application with Noise)

Minm 3 points needed inside the circle.

- The center point is called core point
- The boundary points are the rest 2 points



One cluster's core point
can be another
cluster's boundary point

In circle 1 we have q and r as boundary points and p as core point.

In circle 2 we have p and s as boundary points and r as core point

In circle 3 we cannot call r and s as boundary points or core points as we require 3 points we have only 2.

* To detect outlier we can use dbscan algorithm.

To detect outliers for supervised learning we have graphs
lmplot, boxplot, scatterplot

To detect outlier for unstructure data we use dbscan algo

* PCA (Principal Component Analysis)

PCA concept will use for 2 important concepts

① overfitting

② cluster

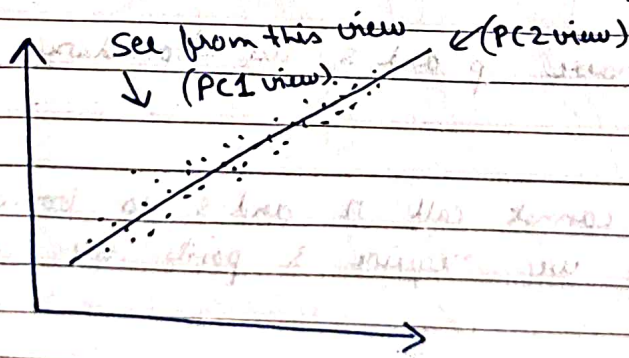
① overfitting

train the model with more attributes then overfitting occurs.

To reduce overfitting we have regularization, cross validation and pca and ensemble learning.

PCA is also called dimension reduction technique.

In PCA converts to low dimension the high dimension by decreasing the number of attributes (less number of attributes).



Using help of PC (Principal Component) which is set of group of datapoint which helps to reduce overfitting.